CURRENT RESEARCH / COMPUTATIONAL ANTIBODY DESIGN

AI-designed biparatopic antibodies against B7-H3

Two distinct epitopes on human CD276 | Goal: same-antigen engagement in a 1A + 1B IgG-Fc

RF2 SCREENING LIBRARY: 1,176 Arm A models + 1,188 Arm B models = 2,364 models

01 TARGET & EPITOPES
Human 4Ig B7-H3
A: IgV1 • B: IgC1

02 DESIGN & SCREEN
RFantibody + ProteinMPNN
RF2 geometry + interface QC

03 ARM VALIDATION
AlphaFold 3
2 seeds × 2 samples per arm

04 DUAL-ARM MODELING
40 Arm A × 29 Arm B
Shared full 4Ig target

05 NEXT: IgG & ASSAYS
Pairing + Fc engineering
Binding, cis engagement, function

69
RF2-shortlisted arm candidates
40 Arm A + 29 Arm B

276
AF3 arm-validation predictions
69 inputs × 4 stochastic samples

1,160
A+B pairwise inputs
164 completed • snapshot: 18 Oct 2026

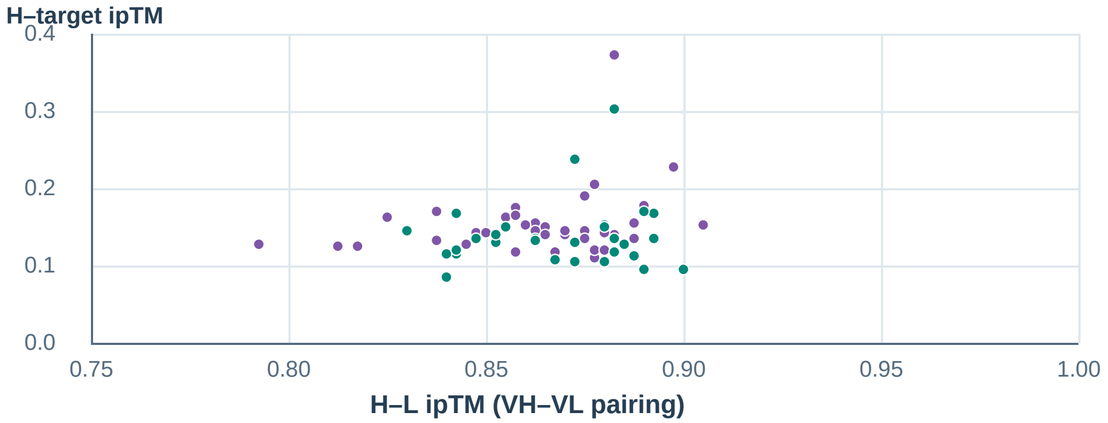
ARM-LEVEL EVIDENCE • Cropped-target AF3 validation cohort; independent of the later full-target pair campaign

● Arm A: IgV1 FG loop (n = 40) ● Arm B: IgC1 patch (n = 29)

Computational shortlist ≠ confirmed binding

Antibody pairing vs. target-interface confidence

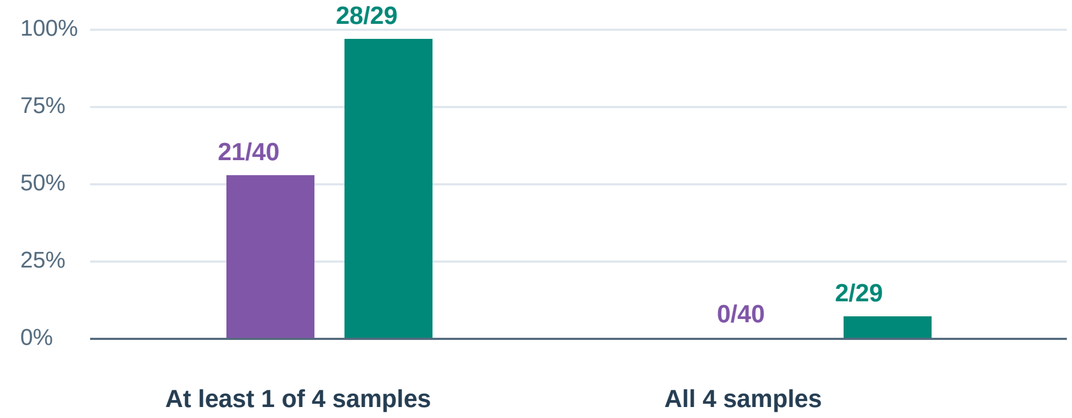
Each point: one arm candidate, mean of four AF3 samples



Strong H–L pairing; low target-interface confidence remains.

Recovery of the selected epitope

≥2 selected residues contacted within 4.5 Å; fraction of inputs



Contact recovery is variable; it does not establish affinity.

NEXT MILESTONE: validate simultaneous A+B binding on one B7-H3 molecule, assemble IgG-Fc and test experimentally.

Pipeline: RFantibody (RFdiffusion) → ProteinMPNN → RF2 → AlphaFold 3 | Recorded results through 18 October 2026